ATISHNA SAMANTARAY

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AI researcher pursuing a Ph.D. in Electrical and Computer Engineering at UC San Diego (advisor: Prof. Pamela Cosman), focused on generative and diffusion models, multimodal video synthesis, and cross-view representation learning. Experienced designing and validating models end-to-end: from architecture and training to synthetic-data generation and rigorous human/model evaluation, with work spanning multimodal video generation, perceptual image/video compression, 3D perception on autonomous-driving sensor data, and medical imaging.

RESEARCH INTERESTS

Image and video compression · vision models · generative models · generative AI · multimodal learning · video generation

EDUCATION

Ph.D., Electrical and Computer Engineering *University of California, San Diego* Sep 2024 – Expected Jun 2028

M.S., Electrical and Computer Engineering *University of California, San Diego* Sep 2022 – Jun 2024

B.Tech., Electronics and Telecommunication Engineering *IIEST, Shibpur* Aug 2018 – Jun 2022

SELECTED PUBLICATIONS & MANUSCRIPTS

Cross-View Attention and Diffusion Synthesis for Maxillary Sinus Pathology Assessment - **A. Samantaray**, X. Cai, K. Archambault, P. Cosman. *Submitted to IEEE TMI.*

Attention Where It Matters: Implicit Saliency Priors in Learned Image Compression - **A. Samantaray**, T. Homavazir, G. Mundada, S. Kamate, P. Cosman. *Submitted to IEEE TIP.*

Can We Remove the Ground? Obstacle-Aware Point Cloud Compression for Remote Object Detection - P. Zheng, A. Presta, **A. Samantaray**, J. Reinis, D. Bharadia, H. Qiu, P. Cosman. *ICNC 2026.*

Novel Algorithm for COVID-19 Variant Detection from Chest X-Rays - **A. Samantaray**, S. Bandhyopadhyay, A. Pramanik. *De Gruyter, 2023.*

Fast Trilateral Filtering for Video Denoising - **A. Samantaray**, S. Bhattacharya. *IEEE ICCCA 2020.*

RESEARCH EXPERIENCE

Graduate Student Researcher *University of California, San Diego* Sep 2024 – Present

Minimal-Signal Egocentric Reconstruction

- Designing a **multimodal video synthesis framework** to reconstruct egocentric video from a compact signal set (sparse exo/ego keyframes, head pose, gaze, hand-pose trajectories, HOI masks) for **AR/VR skill-transfer applications**.
- Formulating an **event-driven anchor-refresh and bitrate allocation policy** across modalities to **reduce storage/transmission bandwidth** while preserving reconstruction and downstream action-recognition quality.

Cross-View Attention and Diffusion Synthesis for Maxillary Sinus Pathology Assessment
(submitted, IEEE TMI)

- Built a **structure-guided latent diffusion model** (**Stable Diffusion + LoRA**, **frozen VAE**, DDIM sampling) that adapts to the CBCT domain, generating an expert-validated synthetic pathology dataset to address severe class imbalance across 194 patients.
- Improved maxillary sinus disease classification accuracy by up to 10.8% by designing a **bidirectional cross-view attention module** across three backbone architectures (**CNN+ViT, YOLOv5, ResNet18**).
- Evaluated synthesis quality with KID scores (**InceptionV3, RadImageNet**) and UMAP visualization, with radiologist verification of generated pathologies.

Attention Where It Matters: Implicit Saliency Priors in Learned Image Compression *(submitted, IEEE TIP)*

- Achieved **15–20% improvements** in perceptual image quality metrics (LPIPS, DISTS) and a **1.9x higher human preference rate** over standard learned-compression baselines by designing **Saliency-Driven Attention (SDA)**, a training-only mechanism injecting human gaze maps as a soft multiplicative prior into transformer self-attention.
- Applied SDA across **three transformer architectures (STF, TCM, SwinT)** and validated results via a 30-participant human study (216 image pairs/participant), with SDA reconstructions preferred in **50.6% of trials vs. 31.6%** for baselines (**$p < 0.001$**).

Graduate Student Researcher *University of California, San Diego* Sep 2022 – Sep 2023

Obstacle-Aware Point Cloud Compression for Remote Object Detection *("Can We Remove the Ground?," ICNC 2026)*

- Designed a lightweight, highly parallelizable **Pillar-based Ground Removal (PGR) algorithm** that selectively removes non-informative LiDAR ground points, cutting point cloud bitrate while preserving 3D object detection accuracy on the Waymo Open Dataset and KITTI.
- Achieved **real-time performance (86.5 FPS)** integrating PGR with SOTA 3D object detectors (SECOND, PVT-SSD), reducing bit rate by up to **12.94%** with negligible mAP degradation across Vehicle, Pedestrian, and Cyclist classes.

TEACHING EXPERIENCE

- Teaching Assistant, Advanced Data Structures and Algorithms - Spring 2023
- Teaching Assistant, Applications of Digital Signal Processing - Spring 2024

TECHNICAL SKILLS

Languages: Python, C, C++, MATLAB, R

Frameworks & Tools: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, NumPy, SciPy, Pandas, Matplotlib, AWS (Cloud Computing), Docker, Git

Modeling & Techniques: Diffusion Models (Stable Diffusion, LoRA fine-tuning), Transformers, CNNs, Generative Models, Cross-Modal / Cross-View Representation Learning, Multimodal Learning, Point Cloud Processing, multimodal understanding, human evaluation, visual quality assessment

Subject Knowledge: Machine Learning, Image & Video Compression, Video Understanding, Computer Vision, LLMs, VLMs, Statistical Learning, Human-Study Design & Evaluation, Image Denoising, Image super-resolution, Image reconstruction, Embedded Systems, GPU, 3D Perception

Coursework: Digital Image Processing, Digital Signal Processing, Probability & Statistics, Statistical Learning, Linear Algebra